| Question | Answer | Marks | AOs | | Guidance |
|---|---|---|---|---|---|
| 1 | **EITHER**<br>$f(2)=3\times2^3-8\times2^2+3\times2+2=24-32+6+2=0$<br><br>Therefore by the factor theorem $(x-2)$ is a factor | **M1**<br>**A1**<br><br>**E1**<br>**[3]** | **1.1a**<br>**1.1b**<br><br>**2.2a** | **AG**<br>Function notation need not be used<br>Zero must be seen<br><br>Reason required | |
| | **OR**<br>$f(x)=(x-2)(3x^2-2x-1)$<br><br>No remainder so $(x-2)$ is a factor | **M1**<br><br>**A1**<br>**E1**<br>**[3]** | | Using algebraic division as far as $3x^2$<br>Correct quotient<br>Reason required | |